

Transformers and Attention Note-Taking Framework

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1 Summary

This weekend we looked at Transformers and Attention. Transformers were introduced as a new architecture that could be used instead of the existing methods such as recurrent neural networks, Long Short Term Memory, etc. This new architecture relies entirely upon self-attention. Self-attention is used to relate different positions of tokens in a sentence to find a representation of the sentence (Attention is All You Need, 1-2).

Transformers are composed of two main parts, an encoder and a decoder. The encoder takes in input symbols and assigns them to a continuous representation. Then the decoder takes this continuous representation and generates the output symbols. Both the encoder and decoder are made up of a stack of identical layers. Each layer has two sub-layers, self-attention and positional encoding. The decoder has an additional sub-layer for encoder-decoder attention.

The self-attention sub-layer compares the input word to all the other input words, including itself. Using the input word's embedding we calculate query, key, and value vectors. A benefit of this is that each of these vectors can be calculated at the same time for all input words, this allows the Transformer model to operate in parallel which speeds up training time. We compare the query to the key of other inputs and use softmax to determine how much one word will effect the other's self-attention value (Transformer Neural Networks, Chat-GPT's Foundation, Clearly explained!!!, 13:00-20:00). The Positional Encoding sub-layer uses sin and cosine functions to add positional information to the word embeddings for each input word. The position value for each input word is the function value at each vector index (Transformer Neural Networks, Chat-GPT's Foundation, Clearly explained!!!, 7:30-12:00). The position values and self-attention values are added to the embedding of each word to get the Residual Connection Values for each input word. The additional sub-layer for the decoder is a similar process as the self-attention sub-layer, the main difference being that instead of looking at input words it compares Residual Connection Values from the encoder to the Residual Connection Values from the decoder. It then adds the encoder-decoder attention vector for each output word to its embedding vector.

The final step is to use a Fully Connected Layer, a simple neural network with weights and biases, to select the correct output tokens. We get the output values from the neural network and then run them through a final softmax function to select the output word/token. This process continues until the model gets an EOS (end of sentence) token (Transformer Neural Networks, ChatGPT's Foundation, Clearly explained!!!, 32:00-33:00).

2 Attention Grabbers

1. Right away, it jumped out to me in the paper that they literally created an entirely new architecture. Not quite from scratch, but nothing like it had ever been done before.
2. Another thing in the paper that I thought was interesting was the training efficiency of the transformer architecture. Being able to do the self-attention in parallel is huge, and it trained significantly faster than the best methods of the time.
3. I think it's really interesting just how well this brand new method performed compared to the current state-of-the-art approaches. It made me think of how psyched these people must have been to see their results, they opened a whole new door in the field.
4. In the video, he talks about how back propagation works using a graph and slope as an example. I really like this visual, it made it make more sense. It's hard to think about how it does that with weights in a network, but seeing it with something simple like slope on a graph made the concept simpler to understand if that makes sense.
5. I could be wrong, but I think the video was wrong about how many layers of self-attention the paper used. In the video he said they used 8 layers, but in the paper I'm pretty confident they said they used 6 identical layers for the encoder and decoder respectively.

3 Questions

1. The paper talks about the model being auto-regressive, it can only depend on known outputs that came before the current position. Is there a way to do this from both sides, where the current position could look at positions both before and after it?
2. Was there a specific thing that sparked the idea behind the paper and using just attention? Or did it just happen to be something that was tried?

3. In both the paper and video transformers are used to translate a sentence from one language to another. What other tasks are transformers good at?
4. Do you need to have a separate input and output vocabulary when working with transformers?
5. What is the BLEU score and what does it rank?